IT2140- Database Design and Development

Year 2, Semester 1

Lab Sheet 03: Creating Tables with Constraints in SQL

Objective

At the end of this lab session, you will be able to use the CREATE TABLE command to create database tables and specify Primary Key, Foreign Key, and Check constraints on the tables created.

Section 01: Understanding CREATE TABLE and Constraints

1. Creating Tables

The CREATE TABLE statement is used to create a new table in a database. This is the first step when designing a database schema. You define the table structure by specifying columns and their data types.

Syntax:

```
CREATE TABLE <table_name> (
    <column_name1> data_type,
    <column_name2> data_type,
    <column_name3> data_type,
    ...
);
```

Explanation:

You replace <table_name> with the name of your table and list each column with its name and data type. The data type defines what kind of data can be stored in that column (e.g., numbers, text, dates).

2. Common Data Types in SQL

Data type	Definition
Integer	Used to store whole numbers (e.g., 1, 10, 100)
Char()	Fixed length text; stores exactly n characters. Useful for codes or IDs (e.g., NIC number).
Varchar()	Variable-length text up to n characters. Suitable for names, addresses.
Datetime	Stores date and time values, like '2025-07-08 14:00:00'.
Real	Stores decimal numbers (floating point).
Money	Stores currency values with decimal precision.

Explanation:
Choosing the correct data type is important for data accuracy, storage efficiency, and query performance.

3. Constraints in SQL

Constraints are rules applied to table columns to ensure data integrity and enforce relationships.

Constraint	Syntax	Definition
Primary Key Constraint	**1. PRIMARY KEY (COLUMN_NAME)** **2. CONSTRAINT <CONSTRAINT_NAME> PRIMARY KEY (COLUMN_NAME)**	Uniquely identifies each record in a table. Values must be **unique** and **not null**. Each table can have only **one primary key**.
Foreign Key Constraint	**1. FOREIGN KEY (referencing_column) REFERENCES <referenced_table> (referenced_column)** **2. CONSTRAINT <constraint_name> FOREIGN KEY (referencing_column) REFERENCES <referenced_table> (referenced_column)**	Used to link two tables together. The foreign key in one table refers to the **primary key** in another. Enforces **referential integrity**.
Check Constraint	**1. CHECK (condition)** **2. CONSTRAINT <constraint_name> CHECK (condition)**	Limits the values that can be entered in a column based on a condition. Ensures data validity by restricting inputs.

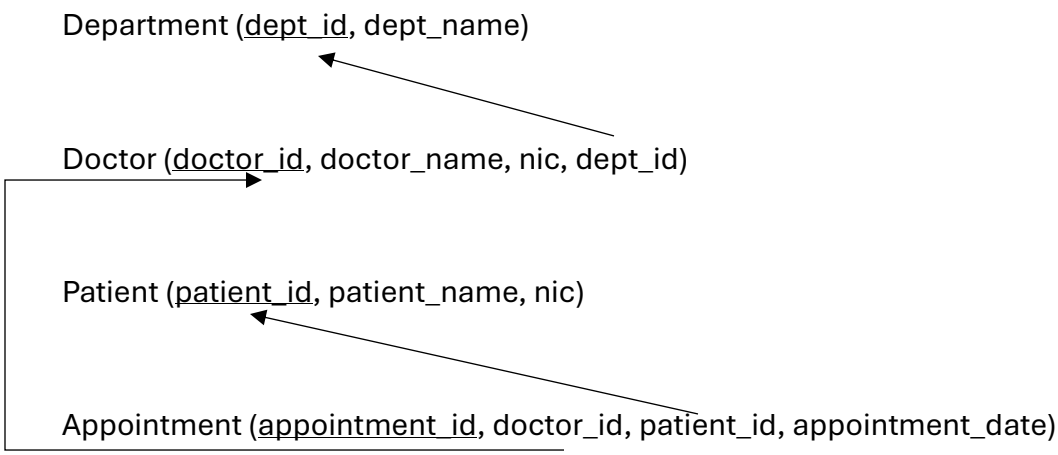
Tips for Students:

While both syntax options work, using the CONSTRAINT <constraint_name> form (option 2) is preferred because naming constraints helps easily identify which constraint was violated when an error occurs.

Section 02: Practical Exercises

1. Identify Primary Keys and Foreign Keys

   Based on the given diagram, fill in the primary and foreign keys for each table.



   Tip for Students:

   In the given schema, the **head of the arrow** indicates the **Primary Key (PK)**, while the **tail of the arrow** indicates the **Foreign Key (FK).**

Table Name	Primary Key(PK)	Foreign Key(FK)
Department		
Doctor		
Patient		
Appointment		

2. Table Creation Order

Determine the correct order to create tables.

Explanation:
When creating tables, those that do not have foreign keys must be created first because tables with foreign keys depend on the referenced tables. Creating tables in the wrong order will result in errors due to missing referenced tables.

3. Create Tables with Constraints

   Write the CREATE TABLE statements for the above schema. Include:

   - Appropriate data types for each column.
   - Primary key constraints to uniquely identify records.
   - Foreign key constraints to maintain referential integrity between tables.

   Explanation:
   This exercise will help you apply your knowledge to design tables that represent a real-world relational database with enforced data rules.

4. Difference Between ALTER and DROP in SQL

   - **ALTER:** Used to modify an existing table's structure, such as adding or dropping columns, or adding constraints.
   - **DROP:** Used to delete an entire table and all its data permanently from the database.
   - **Explanation:** Understanding these commands helps in maintaining and managing database schemas over time.

5. Adding CHECK Constraints Using ALTER TABLE

   Modify the existing Doctor and Patient tables to enforce rules on the data using the CHECK constraint.

   - **Rules:**
     - The Doctor's NIC number must be exactly 9 digits followed by a letter 'V' or 'v'.

```
ALTER TABLE Doctor

ADD CONSTRAINT chk_doctor_nic CHECK (doctor_nic LIKE '[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][Vv]'
```

```
ALTER TABLE Patient

ADD CONSTRAINT chk_patient_nic CHECK (patient_nic LIKE '[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][Vv]'
```

- Appointment date must be in the future.

```
ALTER TABLE Appointment

ADD CONSTRAINT chk_appointment_date CHECK
(appointment_date >= GETDATE())
```

Explanation:

The CHECK constraint restricts the allowed data values and helps maintain data quality.